

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.067 / 0.0186815
Time	2026-08-02 18:00 UTC
Location	29.6239, -95.4742
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	8 / 779	20.95 to 100; mean 68.8698	34.21 at 2026-08-02 17:55Z; dt 5 min
asos	temperature	degC	8 / 779	23.8889 to 37; mean 30.3363	34 at 2026-08-02 17:55Z; dt 5 min
asos	wind_direction	degree	8 / 1026	0 to 360; mean 180.984	320 at 2026-08-02 17:55Z; dt 5 min
asos	wind_speed	m/s	8 / 1061	0 to 33.4389; mean 2.9242	2.57222 at 2026-08-02 17:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 143	5875.65 to 5947.25; mean 5902.29	5888.01 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	pbl_height	m	11 / 143	31.9203 to 2539.72; mean 835.619	2407.74 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	precipitable_water	mm	11 / 143	26.7445 to 57.7464; mean 39.4324	34.901 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	surface_pressure	hPa	11 / 143	1001.37 to 1013.29; mean 1008.24	1008.5 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	t_850	degC	11 / 143	18.4016 to 22.5646; mean 20.3009	19.9148 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	u_10m	m/s	11 / 143	-3.62395 to 4.5116; mean 0.9894	-1.15038 at 2026-08-02 18:00Z; dt 0 min
noaa_gfs	v_10m	m/s	11 / 143	-3.70918 to 6.32564; mean 1.1082	-2.76918 at 2026-08-02 18:00Z; dt 0 min
openaq	ozone	ppm	16 / 1039	0 to 0.081; mean 0.029	0.067 at 2026-08-02 18:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 217	11 to 153; mean 28.4009	14 at 2026-08-02 18:00Z; dt 0 min
openaq	pm25	ug/m3	71 / 2786	0 to 50.8; mean 8.1906	7.7 at 2026-08-02 18:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 27.025	96 at 2026-08-02 18:01Z; dt 2 min
openweather	humidity	percent	5 / 360	34 to 96; mean 70.0806	47 at 2026-08-02 18:01Z; dt 2 min
openweather	precipitation	mm/h	5 / 360	0 to 0.22; mean 0.0009	0 at 2026-08-02 18:01Z; dt 2 min
openweather	pressure	hPa	5 / 360	1007 to 1014; mean 1010.8	1011 at 2026-08-02 18:01Z; dt 2 min
openweather	temperature	degC	5 / 360	25.3 to 37.65; mean 31.1703	35.32 at 2026-08-02 18:01Z; dt 2 min
openweather	wind_direction	degree	5 / 360	0 to 355; mean 182.858	343 at 2026-08-02 18:01Z; dt 2 min
openweather	wind_speed	m/s	5 / 360	0.26 to 5.85; mean 2.1516	3.11 at 2026-08-02 18:01Z; dt 2 min
purpleair	pm25	ug/m3	73 / 5150	0 to 313; mean 10.5459	12.5 at 2026-08-02 18:00Z; dt 0 min
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_ch4_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_co_column	mol/m^2	8 / 8	0.0275078 to 0.0328455; mean 0.0306	0.0328455 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000197921 to 0.000318768; mean 0.0003	0.000306598 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	4.01789e-05 to 5.22023e-05; mean 0	4.15036e-05 at 2026-08-02 19:10Z; dt 70.3 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-08-02 19:10Z; dt 70.3 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-6.07222e-05 to 0.000178122; mean 0	-6.07222e-05 at 2026-08-02 18:38Z; dt 38.9 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; dt 38.9 min
tceq	co	ppm	3 / 188	0 to 0.7; mean 0.1957	0.1 at 2026-08-02 17:00Z; dt 60 min
tceq	no2	ppb	12 / 735	-2.4 to 33.6; mean 4.5578	2 at 2026-08-02 18:00Z; dt 0 min
tceq	so2	ppb	3 / 187	-0.6 to 7.2; mean -0.0214	-0.2 at 2026-08-02 18:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	08-01 06Z 92.17, 12Z 74.32, 18Z 49.89, 08-02 00Z 75.74, 06Z 87.39, 12Z 61.93, 18Z 36.74, 08-03 00Z 70.06, 06Z 88.9, 12Z 68.67, 18Z 46.4, 08-04 00Z 75.82
asos	temperature	08-01 06Z 25.84, 12Z 30.26, 18Z 34.18, 08-02 00Z 30.04, 06Z 27.87, 12Z 31.06, 18Z 35.13, 08-03 00Z 29.49, 06Z 26.25, 12Z 30.64, 18Z 34.42, 08-04 00Z 28.5
asos	wind_direction	08-01 06Z 170.9, 12Z 235.2, 18Z 233.5, 08-02 00Z 211.6, 06Z 237.4, 12Z 127.4, 18Z 179.2, 08-03 00Z 164.5, 06Z 140.4, 12Z 158.8, 18Z 155.6, 08-04 00Z 160.5
asos	wind_speed	08-01 06Z 2.194, 12Z 4.162, 18Z 3.35, 08-02 00Z 3.058, 06Z 2.814, 12Z 1.961, 18Z 2.919, 08-03 00Z 2.942, 06Z 1.526, 12Z 2.166, 18Z 4.485, 08-04 00Z 3.486
noaa_gfs	gh_500	08-01 06Z 5945, 12Z 5927, 18Z 5927, 08-02 00Z 5905, 06Z 5901, 12Z 5881, 18Z 5887, 08-03 00Z 5877, 06Z 5888, 12Z 5881, 18Z 5896, 08-04 00Z 5900, 06Z 5914
noaa_gfs	pbl_height	08-01 06Z 412.6, 12Z 369.7, 18Z 1640, 08-02 00Z 947.7, 06Z 303.3, 12Z 189.3, 18Z 2135, 08-03 00Z 1031, 06Z 184, 12Z 96.21, 18Z 2243, 08-04 00Z 997.7, 06Z 314.1
noaa_gfs	precipitable_water	08-01 06Z 38.49, 12Z 42.19, 18Z 50.15, 08-02 00Z 56.45, 06Z 51.47, 12Z 40.43, 18Z 32.12, 08-03 00Z 29.75, 06Z 31.65, 12Z 32.87, 18Z 34.35, 08-04 00Z 39.01, 06Z 33.68
noaa_gfs	surface_pressure	08-01 06Z 1011, 12Z 1011, 18Z 1009, 08-02 00Z 1005, 06Z 1008, 12Z 1008, 18Z 1008, 08-03 00Z 1005, 06Z 1008, 12Z 1009, 18Z 1009, 08-04 00Z 1007, 06Z 1009
noaa_gfs	t_850	08-01 06Z 22.22, 12Z 21.93, 18Z 19.69, 08-02 00Z 21.63, 06Z 20.76, 12Z 20.47, 18Z 19.48, 08-03 00Z 20.81, 06Z 19.37, 12Z 18.55, 18Z 18.95, 08-04 00Z 20.01, 06Z 20.03
noaa_gfs	u_10m	08-01 06Z 1.555, 12Z 2.258, 18Z 3.996, 08-02 00Z 1.967, 06Z 2.941, 12Z 1.323, 18Z -0.3522, 08-03 00Z -1.175, 06Z 0.9993, 12Z 1.264, 18Z -0.6977, 08-04 00Z -1.885, 06Z 0.6699
noaa_gfs	v_10m	08-01 06Z 3.258, 12Z 1.821, 18Z 0.6377, 08-02 00Z 0.1537, 06Z 1.568, 12Z -1.663, 18Z -2.897, 08-03 00Z -0.8308, 06Z 2.518, 12Z 0.8871, 18Z 1.824, 08-04 00Z 4.186, 06Z 2.945

Source	Metric	Values
openaq	ozone	08-01 06Z 0.01177, 12Z 0.01407, 18Z 0.03085, 08-02 00Z 0.02182, 06Z 0.009148, 12Z 0.0239, 18Z 0.05994, 08-03 00Z 0.05262, 06Z 0.02483, 12Z 0.0207, 18Z 0.05074, 08-04 00Z 0.0222, 06Z 0.016
openaq	pm10	08-01 06Z 16.94, 12Z 30.22, 18Z 31.83, 08-02 00Z 24.33, 06Z 22.22, 12Z 22.89, 18Z 16.61, 08-03 00Z 20.44, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 38.33
openaq	pm25	08-01 06Z 3.341, 12Z 4.897, 18Z 5.655, 08-02 00Z 8.785, 06Z 9.929, 12Z 8.068, 18Z 8.749, 08-03 00Z 10.32, 06Z 9.128, 12Z 9.872, 18Z 12.05, 08-04 00Z 11.63, 06Z 14.92
openweather	cloud_cover	08-01 06Z 4.8, 12Z 13.1, 18Z 18.45, 08-02 00Z 55.87, 06Z 67.31, 12Z 85.45, 18Z 32.53, 08-03 00Z 19.73, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13
openweather	humidity	08-01 06Z 87.5, 12Z 81.24, 18Z 55.77, 08-02 00Z 71.87, 06Z 85.97, 12Z 70.94, 18Z 41.77, 08-03 00Z 65.2, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7
openweather	precipitation	08-01 06Z 0, 12Z 0, 18Z 0.003871, 08-02 00Z 0, 06Z 0.007586, 12Z 0, 18Z 0, 08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0
openweather	pressure	08-01 06Z 1013, 12Z 1014, 18Z 1010, 08-02 00Z 1009, 06Z 1010, 12Z 1012, 18Z 1009, 08-03 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011
openweather	temperature	08-01 06Z 26.73, 12Z 29.56, 18Z 35.64, 08-02 00Z 31.39, 06Z 28.4, 12Z 30.66, 18Z 36.36, 08-03 00Z 31.37, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17
openweather	wind_direction	08-01 06Z 154.6, 12Z 232.2, 18Z 264, 08-02 00Z 230.7, 06Z 194.6, 12Z 90.61, 18Z 166.6, 08-03 00Z 158.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161
openweather	wind_speed	08-01 06Z 2.446, 12Z 2.695, 18Z 2.235, 08-02 00Z 2.286, 06Z 2.043, 12Z 1.942, 18Z 2.036, 08-03 00Z 1.967, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572
purpleair	pm25	08-01 06Z 4.363, 12Z 6.291, 18Z 7.201, 08-02 00Z 12.48, 06Z 12.2, 12Z 10.37, 18Z 9.796, 08-03 00Z 12.35, 06Z 12.07, 12Z 12.31, 18Z 13.9, 08-04 00Z 13.02
sentinel5p	s5p_aer_ai_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_ch4_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_co_column	08-01 18Z 0.02751, 08-02 18Z 0.03284, 08-03 18Z 0.03095
sentinel5p	s5p_co_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_hcho_column	08-01 18Z 0.0001979, 08-02 18Z 0.0003127, 08-03 18Z 0.0002755
sentinel5p	s5p_hcho_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_no2_column	08-01 18Z 4.018e-05, 08-02 18Z 4.15e-05, 08-03 18Z 4.973e-05
sentinel5p	s5p_no2_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_o3_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	s5p_so2_column	08-01 18Z 6.519e-06, 08-02 18Z -4.61e-05, 08-03 18Z 0.0001056
sentinel5p	s5p_so2_granule_available	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
tceq	co	08-01 06Z 0.1611, 12Z 0.2056, 18Z 0.1812, 08-02 00Z 0.22, 06Z 0.1167, 12Z 0.1333, 18Z 0.16, 08-03 00Z 0.2444, 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.1333
tceq	no2	08-01 06Z 3.42, 12Z 3.552, 18Z 4.337, 08-02 00Z 6.082, 06Z 3.036, 12Z 3.271, 18Z 2.811, 08-03 00Z 4.778, 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 2.683
tceq	so2	08-01 06Z -0.1944, 12Z -0.1167, 18Z -0.04706, 08-02 00Z -0.18, 06Z -0.07778, 12Z -0.09444, 18Z -0.02778, 08-03 00Z -0.03333, 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2667

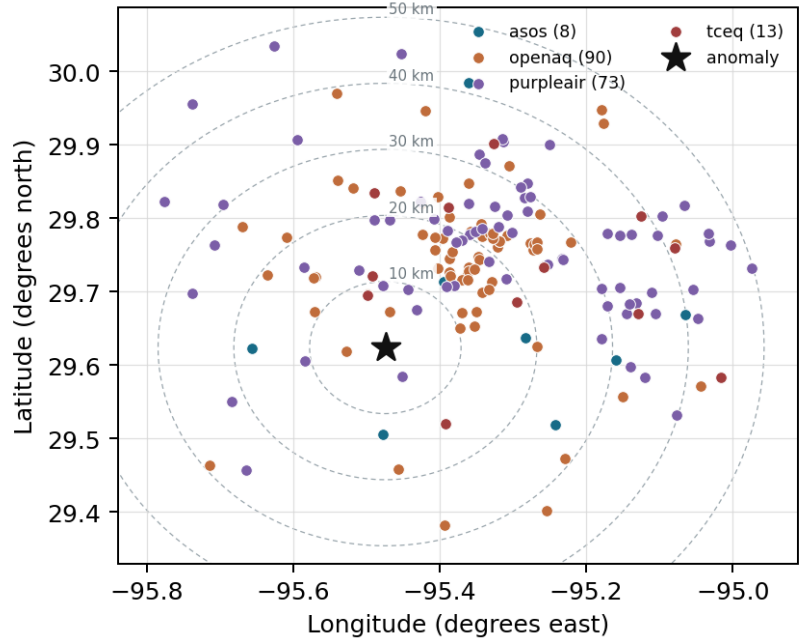
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

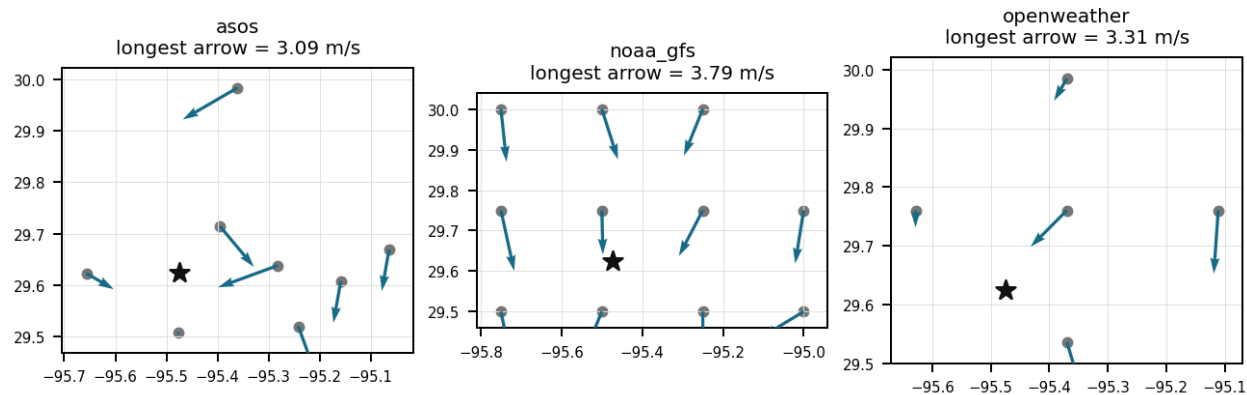
Source	Metric	Values
asos	humidity	MCJ (12.607 km) 66.34, AXH (13.101 km) 68.99, SGR (17.622 km) 73.47, HOU (18.6 km) 69.1, LVJ (25.336 km) 71.43, EFD (30.555 km) 67.37, T41 (39.95 km) 70.28, IAH (41.555 km) 64.7
asos	temperature	MCJ (12.607 km) 30.58, AXH (13.101 km) 29.95, SGR (17.622 km) 29.88, HOU (18.6 km) 30.54, LVJ (25.336 km) 30.14, EFD (30.555 km) 30.84, T41 (39.95 km) 30.17, IAH (41.555 km) 30.99
asos	wind_direction	MCJ (12.607 km) 204.4, AXH (13.101 km) 107.2, SGR (17.622 km) 207, HOU (18.6 km) 205.2, LVJ (25.336 km) 163.9, EFD (30.555 km) 174.5, T41 (39.95 km) 196.8, IAH (41.555 km) 190.3
asos	wind_speed	MCJ (12.607 km) 3.651, AXH (13.101 km) 1.133, SGR (17.622 km) 2.897, HOU (18.6 km) 3.569, LVJ (25.336 km) 2.19, EFD (30.555 km) 4.29, T41 (39.95 km) 3.454, IAH (41.555 km) 3.04
noaa_gfs	gh_500	gfs:29.50,-95.50 (14.001 km) 5903, gfs:29.75,-95.50 (14.241 km) 5902, gfs:29.50,-95.25 (25.691 km) 5902, gfs:29.75,-95.25 (25.8 km) 5902, gfs:29.50,-95.75 (30.023 km) 5903, gfs:29.75,-95.75 (30.107 km) 5903, gfs:30.00,-95.50 (41.894 km) 5902, gfs:30.00,-95.25 (47.083 km) 5902, +3 more
noaa_gfs	pbl_height	gfs:29.50,-95.50 (14.001 km) 760, gfs:29.75,-95.50 (14.241 km) 1085, gfs:29.50,-95.25 (25.691 km) 703.5, gfs:29.75,-95.25 (25.8 km) 1009, gfs:29.50,-95.75 (30.023 km) 752.8, gfs:29.75,-95.75 (30.107 km) 653.2, gfs:30.00,-95.50 (41.894 km) 1117, gfs:30.00,-95.25 (47.083 km) 958.6, +3 more
noaa_gfs	precipitable_water	gfs:29.50,-95.50 (14.001 km) 40.18, gfs:29.75,-95.50 (14.241 km) 39.54, gfs:29.50,-95.25 (25.691 km) 40.62, gfs:29.75,-95.25 (25.8 km) 39.98, gfs:29.50,-95.75 (30.023 km) 39.79, gfs:29.75,-95.75 (30.107 km) 38.85, gfs:30.00,-95.50 (41.894 km) 38.29, gfs:30.00,-95.25 (47.083 km) 38.82, +3 more
noaa_gfs	surface_pressure	gfs:29.50,-95.50 (14.001 km) 1009, gfs:29.75,-95.50 (14.241 km) 1008, gfs:29.50,-95.25 (25.691 km) 1010, gfs:29.75,-95.25 (25.8 km) 1009, gfs:29.50,-95.75 (30.023 km) 1008, gfs:29.75,-95.75 (30.107 km) 1007, gfs:30.00,-95.50 (41.894 km) 1006, gfs:30.00,-95.25 (47.083 km) 1008, +3 more
noaa_gfs	t_850	gfs:29.50,-95.50 (14.001 km) 20.34, gfs:29.75,-95.50 (14.241 km) 20.3, gfs:29.50,-95.25 (25.691 km) 20.21, gfs:29.75,-95.25 (25.8 km) 20.16, gfs:29.50,-95.75 (30.023 km) 20.45, gfs:29.75,-95.75 (30.107 km) 20.42, gfs:30.00,-95.50 (41.894 km) 20.49, gfs:30.00,-95.25 (47.083 km) 20.25, +3 more
noaa_gfs	u_10m	gfs:29.50,-95.50 (14.001 km) 0.9337, gfs:29.75,-95.50 (14.241 km) 1.098, gfs:29.50,-95.25 (25.691 km) 0.9283, gfs:29.75,-95.25 (25.8 km) 0.7845, gfs:29.50,-95.75 (30.023 km) 1.384, gfs:29.75,-95.75 (30.107 km) 1.306, gfs:30.00,-95.50 (41.894 km) 1.032, gfs:30.00,-95.25 (47.083 km) 0.3429, +3 more
noaa_gfs	v_10m	gfs:29.50,-95.50 (14.001 km) 1.345, gfs:29.75,-95.50 (14.241 km) 1.067, gfs:29.50,-95.25 (25.691 km) 1.656, gfs:29.75,-95.25 (25.8 km) 1.06, gfs:29.50,-95.75 (30.023 km) 0.9803, gfs:29.75,-95.75 (30.107 km) 0.788, gfs:30.00,-95.50 (41.894 km) 0.8473, gfs:30.00,-95.25 (47.083 km) 0.688, +3 more
openaq	ozone	312 (0.0 km) 0.0303, 311 (8.352 km) 0.03034, 3055 (13.969 km) 0.02859, 1319333 (18.219 km) 0.03497, 325 (18.683 km) 0.0125, 287 (19.129 km) 0.03039, 320 (20.009 km) 0.0269, 319 (23.429 km) 0.03007, +8 more
openaq	pm10	15852419 (22.1 km) 18.48, 13380082 (24.246 km) 41.51, 271 (33.799 km) 25.13
openaq	pm25	15399615 (5.18 km) 7.362, 8967119 (5.464 km) 9.038, 8967021 (10.251 km) 8.895, 13955816 (10.822 km) 8.504, 9416627 (11.449 km) 5.518, 15472101 (12.224 km) 5.806, 13418566 (13.238 km) 7.463, 8966939 (13.831 km) 8.489, +63 more
openweather	cloud_cover	grid:south (14.067 km) 31.94, grid:center (18.223 km) 26.07, grid:west (21.273 km) 28.04, grid:east (38.217 km) 27.28, grid:north (41.397 km) 21.79
openweather	humidity	grid:south (14.067 km) 70.89, grid:center (18.223 km) 68.6, grid:west (21.273 km) 69.46, grid:east (38.217 km) 74.01, grid:north (41.397 km) 67.44
openweather	precipitation	grid:south (14.067 km) 0, grid:center (18.223 km) 0, grid:west (21.273 km) 0.001667, grid:east (38.217 km) 0.003056, grid:north (41.397 km) 0
openweather	pressure	grid:south (14.067 km) 1011, grid:center (18.223 km) 1011, grid:west (21.273 km) 1011, grid:east (38.217 km) 1011, grid:north (41.397 km) 1011
openweather	temperature	grid:south (14.067 km) 30.95, grid:center (18.223 km) 31.65, grid:west (21.273 km) 31.14, grid:east (38.217 km) 30.86, grid:north (41.397 km) 31.26
openweather	wind_direction	grid:south (14.067 km) 213.3, grid:center (18.223 km) 200, grid:west (21.273 km) 189.7, grid:east (38.217 km) 172.3, grid:north (41.397 km) 139
openweather	wind_speed	grid:south (14.067 km) 2.159, grid:center (18.223 km) 2.077, grid:west (21.273 km) 1.383, grid:east (38.217 km) 3.103, grid:north (41.397 km) 2.037
purpleair	pm25	161003 (4.842 km) 5.34, 166451 (7.009 km) 10.42, 46237 (9.253 km) 0.3681, 186315 (9.433 km) 11.82, 276512 (10.831 km) 11.63, 319092 (12.274 km) 15.48, 293735 (12.3 km) 12.05, 293725 (13.021 km) 12.37, +65 more

Source	Metric	Values
tceq	co	482011052 (22.765 km) 0.2703, 482011035 (24.228 km) 0.1667, 482011039 (33.792 km) 0.1475
tceq	no2	482010055 (8.347 km) 2.61, 482011066 (11.011 km) 7.859, 480391004 (13.954 km) 1.683, 482010416 (18.679 km) 4.703, 482011052 (22.765 km) 9.068, 482010047 (23.429 km) 3.44, 482011035 (24.228 km) 7.105, 482011039 (33.792 km) 3.05, +4 more
tceq	so2	482010051 (0.017 km) -0.2492, 482011035 (24.228 km) -0.2825, 482011039 (33.792 km) 0.4836

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The PM_{2.5} levels exceeded the normal range in Houston, Texas on August 2nd, indicating a significant increase in particulate matter in the air.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 2 of 36

The wind direction and speed are conducive to trapping pollutants in the Houston area, exacerbating the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 3 of 36

Increased volatile organic compound activity, indicated by elevated satellite formaldehyde column amounts around 0.00031 mol/m², fueled atmospheric ozone formation in the area.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 4 of 36

West-central Houston ozone at 2026-08-02 18:00 UTC is most consistent with an aged ozone plume rather than a fresh local combustion plume because ozone reached 0.067 ppm while nearby NO₂ was only 2.0 ppb, TCEQ CO was low at 0.1 ppm within the hour, and nearby SO₂ was not elevated.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 5 of 36

Sentinel-5P satellite measurements revealed elevated total column formaldehyde concentrations on August 2 relative to August 1, indicating increased volatile organic compound reactivity that promoted rapid daytime ozone production.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 6 of 36

The pollutant involved is particulate matter (PM_{2.5}), which is elevated to 12.5 ug/m³.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 7 of 36

The anomalous pollutant was ozone measured by openaq at 29.624, -95.474 in Houston, Texas, at 2026-08-02T18:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 8 of 36

VOC-sensitive photochemistry from urban-industrial emissions likely contributed to the ozone spike, because elevated formaldehyde and carbon monoxide columns coincided with the event while nearby NO₂ and SO₂ were low, a pattern consistent with an aged, oxidized ozone-producing air mass rather than a fresh combustion plume.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 9 of 36

The observed ozone value at the anomalous location and time was 0.067 ppm, compared with an expected value of 0.018681528662420382 ppm, corresponding to a z-score of 5.6520258945453925 and severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 10 of 36

The 6-hour mean ozone concentration from openaq for 2026-08-02 18Z was 0.05994 ppm, which was higher than the adjacent 6-hour means of 0.0239 ppm at 2026-08-02 12Z and 0.05262 ppm at 2026-08-03 00Z within the provided context window.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 11 of 36

The temperature is unusually high, causing increased emissions from industrial and vehicular sources.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 12 of 36

Surface ozone concentrations in Houston reached a severe peak of 0.067 ppm at 18:00 UTC on August 2, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 13 of 36

A deep and well-mixed boundary layer supports efficient afternoon ozone buildup and vertical mixing near west-central Houston at 18:00 UTC on 2026-08-02 because modeled planetary boundary-layer height reached about 2408 m near the anomaly, far above the low morning values in the same window.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 14 of 36

Hot and dry boundary-layer conditions strongly support enhanced local ozone photochemistry in west-central Houston at 18:00 UTC on 2026-08-02 because surface temperature was about 34 to 35.3 C while relative humidity dropped to about 34 to 47 percent near the anomaly, both consistent with a strongly heated afternoon environment.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 15 of 36

Satellite column measurements over Houston detected elevated formaldehyde column densities of approximately 0.00031 mol/m², indicating active precursor volatile organic compound photochemistry.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 16 of 36

Houston precursor and transport conditions support VOC-rich plume processing and recirculation because satellite HCHO and CO columns were elevated on 2026-08-02 18Z and near-surface winds were weak and variable around the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 17 of 36

The air-quality anomaly in Houston, Texas on August 2nd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 18 of 36

Deep boundary layer expansion exceeding 2,000 meters enabled efficient vertical mixing and buildup of photochemically generated ozone near the surface.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 19 of 36

Weak and shifting winds together with low contemporaneous NO₂ and elevated HCHO support transport of an aged VOC-influenced ozone plume into west-central Houston, although the case is not cleanly attributable to a fresh combustion plume because nearby NO₂ was only about 2 ppb, TCEQ CO was low, and wind direction differed between nearby observations while cloud cover was reported as high at one nearby grid point.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 20 of 36

High surface temperatures exceeding 34 °C combined with low relative humidity below 37% during the afternoon of August 2 created highly favorable atmospheric conditions for photochemical ozone formation.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 21 of 36

A nearby wildfire or agricultural burn has released pollutants into the atmosphere, contributing to the anomaly.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 22 of 36

The anomaly occurred at a location approximately 14.067 km from the query point, with an elevation of 30.95 degrees Celsius and wind direction of 213.3 degrees.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 23 of 36

The air quality in Houston, Texas is anomalous due to high levels of SO₂.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 24 of 36

An ozone anomaly with a concentration of 0.067 ppm was recorded by openaq on 2026-08-02 at 18:00:00 UTC at coordinates (29.624, -95.474), reflecting a z-score of 5.65 above the expected baseline of 0.0187 ppm.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 25 of 36

Weak and shifting low-level winds likely favored recirculation and local accumulation of ozone and precursors over west-central Houston on 2026-08-02, because observed winds were only about 2 to 3 m/s with directional variability and modeled boundary-layer height was very deep by 18Z, supporting inland transport and mixing of regionally produced ozone.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 26 of 36

High daytime surface temperatures reaching up to 35 °C combined with clear conditions and low relative humidity accelerated local photochemical ozone production in Houston around 18:00 UTC on August 2, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 27 of 36

Model data from gfs shows planetary boundary layer heights expanding to between 2135 meters and 2408 meters around 18:00:00 UTC on 2026-08-02 near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 28 of 36

At the time of the ozone anomaly on 2026-08-02 around 18:00:00 UTC, meteorological observations from asos and openweather showed elevated surface temperatures of 34.0 degC to 35.3 degC and relative humidity dropping to between 34.2 percent and 36.7 percent near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 29 of 36

The air quality in Houston, Texas is anomalous due to high levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 30 of 36

The air quality in Houston, Texas is anomalous due to high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 31 of 36

Houston meteorology on 2026-08-02 18:00 UTC strongly favored photochemical ozone formation because surface temperature was about 35 to 36 C, humidity fell to about 34 to 47 percent near the anomaly, and GFS boundary-layer height was about 2408 m at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 32 of 36

The anomaly was detected on August 2nd, 2026, at approximately 18:01:58 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 33 of 36

Hot, unusually dry afternoon conditions near Houston on 2026-08-02 likely accelerated photochemical ozone production, because surface temperature was about 34 to 35.3 C while humidity near the anomaly fell to about 34 to 47 percent at the time of the ozone peak.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 34 of 36

NO₂ levels were also elevated during the anomaly, suggesting increased emissions from industrial or vehicular sources.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 35 of 36

Meteorological observations and forecasts during the anomaly period confirmed high daytime temperatures of 35°C to 36°C, relative humidity near 34% to 37%, and a deep planetary boundary layer height exceeding 2100 meters.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Claim 36 of 36

NOAA GFS model data show planetary boundary layer heights exceeding 2,100 meters at 18:00 UTC, supporting strong convective mixing and efficient vertical distribution of ground-level ozone.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
